FRE-GT-9743 Assignment 2

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xVA

Bond forward with daily VM and funding benefit (computable). Bank A and Client B enter into a bond forward at $t = 0$. Bank A is *long* the forward. The trade is collateralized with *daily variation margin (VM)* in cash under the following rule:

- Let V_t be the end-of-day (EOD) mark-to-market of the forward *from Bank A's perspective* (positive means Bank A is in-the-money).
- The collateral account is updated each day to satisfy $C_t = V_t$ (full collateralization, zero threshold, zero MTA).
- The VM call paid at EOD of day t is $\Delta C_t = C_t - C_{t-1}$.
- Cash collateral posted earns the collateral rate $r_C = 4\%$ (simple interest, ACT/360).

Notional is \$10,000,000 and the forward settlement date is 6 months later (the settlement date is not needed for the calculations below).

At Day 1 EOD, the forward is at par: $V_1 = 0$ (so $C_1 = 0$).

1. On Day 2 EOD, the MTM becomes $V_2 = -200{,}000$. Which party posts VM at Day 2 EOD, and what is the VM amount ΔC_2?
2. Alternatively, suppose on Day 2 EOD the MTM is $V_2 = +150{,}000$. Assume Bank A can invest any received cash at its unsecured funding/investment rate $r_F = 6\%$. Compute the *funding benefit over Day 2* (one day of interest).

Exposure Calculation for a Bond Forward

The key component in calculating xVA is the exposure calculation. Suppose we long a bond forward contract that

- has notional N (e.g., N = 1 mil);
- settles at T (e.g., T = 1Y);
- strikes at K (e.g., K = 100);

At intermediate time $t \in (0, T)$, we shall calculate:

$$E(t) = \max\{V(t), 0\} \tag{1}$$

Here, we assume zero collateralisation, and $V(t)$ is the valuation of bond forward at t.

Q1. In assignment 1, you have worked out the carry method to price a bond forward contract, i.e., basically,

- Repo the spot bond (t=0) to settlement date (T) at r_R;
- Subtract the coupon reinvested at repo rate r_R paid between 0 and T;
- discount the payoff at un-secured funding rate r_F[1];

For exposure, we need to perform the above steps at time t. This requires simulation some relevant dynamics. Claim: we need at least three stochastic factors X_1, X_2, and X_F that enables separating bond yield, repo rate and unsecured funding rate. Give your justification of this claim.

Q2. Let us adopt a multi-factor gaussian model, in which case, a i-discount factor has a closed form in terms of the state variables,

$$P^i(t, T) = \exp\left\{A^i(t, T) - B^i(t, T)^\top \mathbf{X}(\mathbf{t})\right\} \tag{2}$$

and $\mathbf{X}(\mathbf{t})$ follows some known dynamics (Ornstein–Uhlenbeck process), thus A^i and B^i are all known. Is $P^i(t, T)$ known at time t or time T ? Can you express $V(t)$ in terms of $P^i(t, \cdot)$'s ?

Q3. Following the notes, finish the steps to compute CVA for a long position with this contract. (This is just to make sure you understand the procedures to calculate xVA based given exposure profile).

Q4. As a follow up question: how would you come up with a sensible hazard rate that is factored into your survival probability calculation ? (Suppose you're buying from a bank with certain credit rating)

[1] Notice, here we assumed no collateralization, otherwise, it will be discounted at rate r_C.

Sub-Class and Delegation

Part I: Date Utilities

The `date` folder provides a set of utility functions related to date handling. You are expected to familiarize these date-related functions and may test or debug them using the provided test code.

No code implementation is required in this part.

Part II: Linear Products

The code templates for **Bullet Cashflow** and **Overnight Index** will be provided. Students are required to complete the implementations for **Interest Rate Stream** and **Swap**.

After completing the required code, you should be able to display the full information of the corresponding financial products.

If you are not familiar with swap, please review the relevant past lectures on **Notion** before starting this part.

Part III: Option Strategy Registry

Please use the singleton class to finish the registry of the OptionStrategyRegistry.